ECE332 Lab 1: Signal Integrity Lab

Term: Spring 2026

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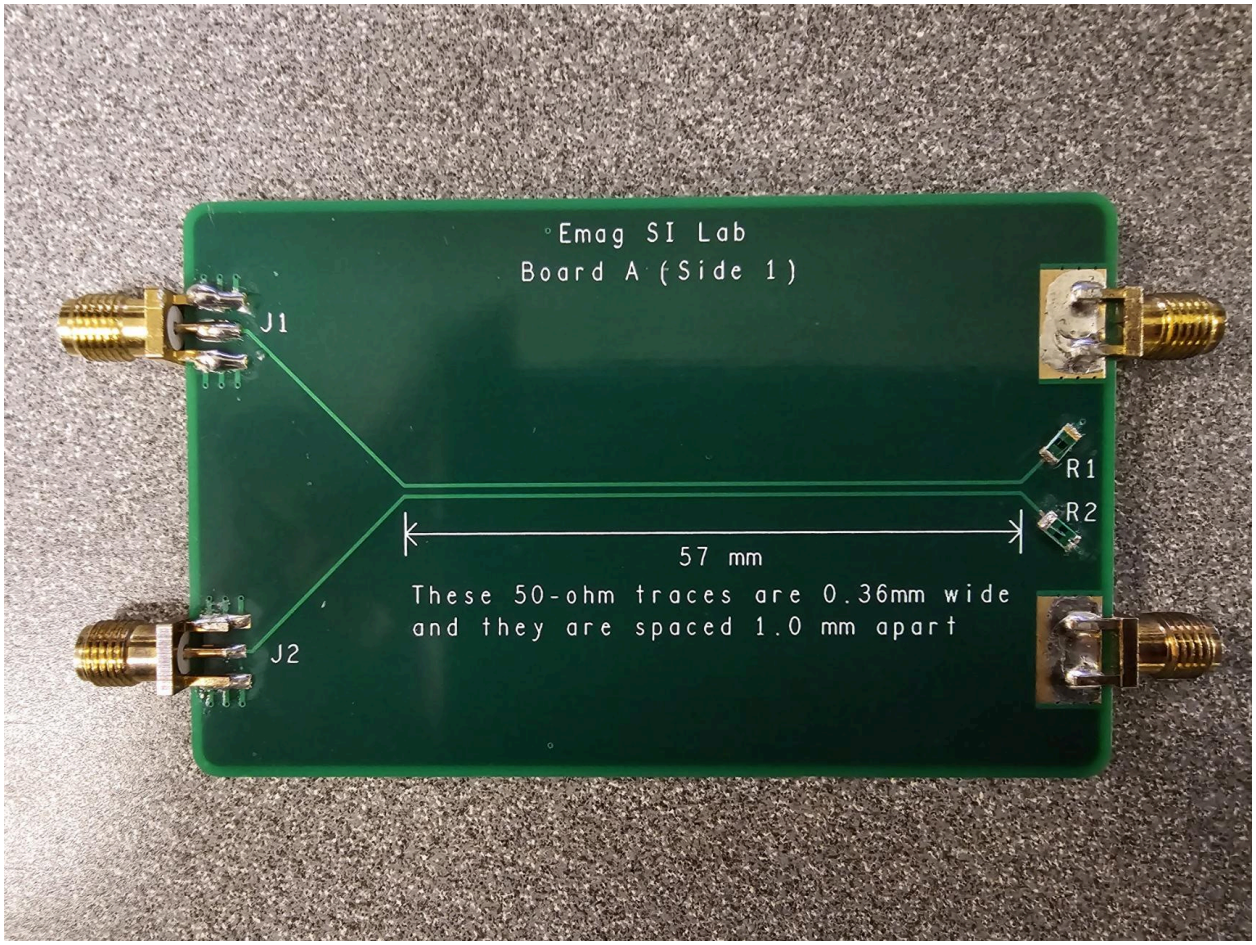
Part 1 – Board A (Side 1)



Objective

The goal of this part is to measure how much two nearby signal traces interfere with each other (coupling) and to check if the traces are properly matched to 50 Ω using a Vector Network Analyzer (VNA).

Theory

Signal integrity focuses on maintaining clean signals and minimizing interference. When two traces are close together, their electric fields interact, which causes coupling.

- **S21** represents how much signal is transmitted between two ports (coupling)
- **S11** represents how much signal is reflected (impedance matching)

Good signal integrity means:

- Low coupling (S21 is very negative)
- Good matching (S11 is less than –10 dB)

Signals travel as electromagnetic waves along transmission lines. If there is an impedance mismatch, part of the wave reflects back, which affects both S11 and S21.

Setup

The VNA was powered on, and the eCal module was connected and allowed to warm up. The display was configured to show three traces: a Smith Chart displaying S11 on the left, and log magnitude plots of S11 and S21 on the right.

The frequency range was set from 10 MHz to 1 GHz. Two SMA cables were connected to the VNA ports, and the log magnitude scale was adjusted to 9 divisions. The VNA was then calibrated using the eCal module.

After calibration, Port 1 was connected to J1 and Port 2 was connected to J2 on Board A. The S21 plot was zoomed in, and trace markers were used to identify the highest and lowest points. The display was then returned to the full three-plot view to analyze S11.

Equations used in each experiment

No calculations were required. Measurements were taken directly using:

- S21 (transmission/coupling)
- S11 (return loss)

Results/Data

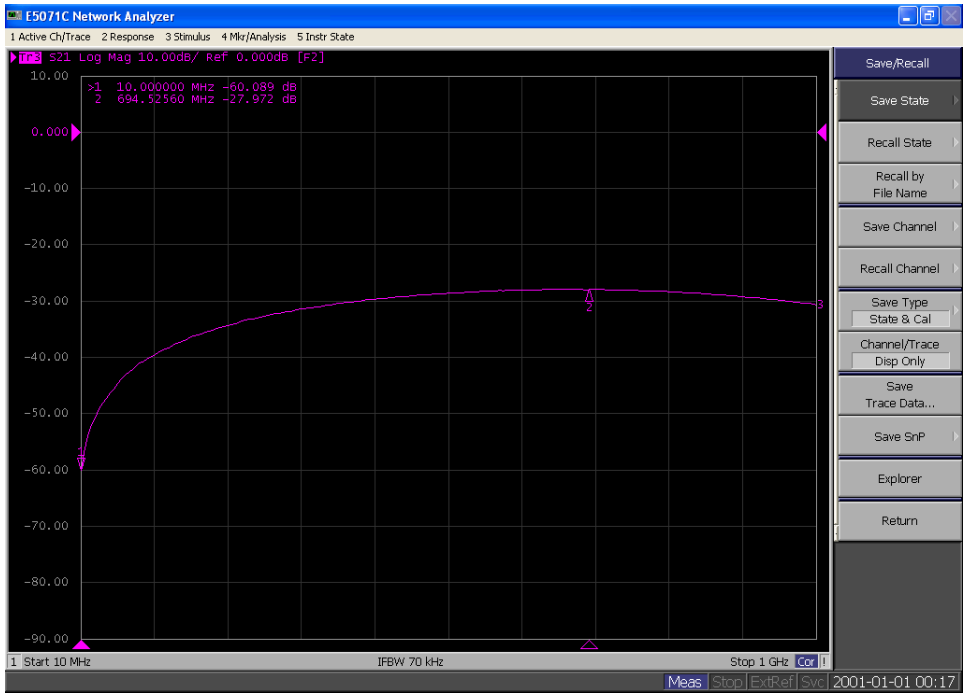


Figure 1. Zoomed S21 plot of Board A, Side 1.

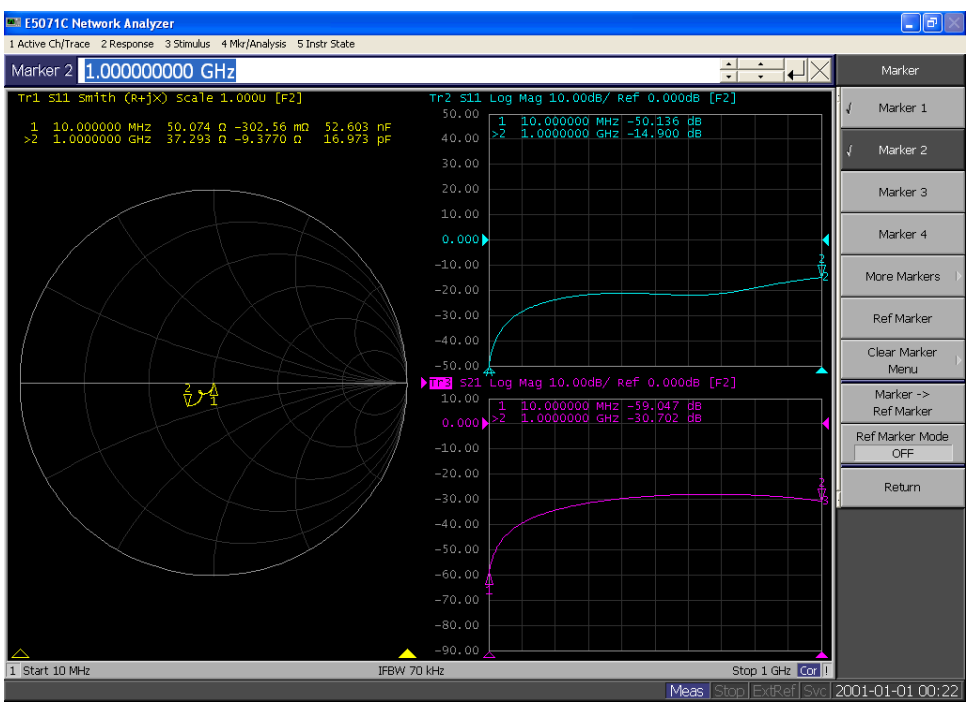


Figure 2. Board A, Side 1 showing S11 and S21 plots.

Observed:

- S21 ranged from about –60 dB to –28 dB
- S11 remained below –10 dB across most frequencies

Discussion

The S21 plot shows how much signal is coupling between the traces. Since the values are very negative, this indicates low coupling, which is desired for independent signals.

If these were two independent signals, we would want low coupling to avoid interference. If they were part of a power transfer system such as a transformer, higher coupling would be desirable.

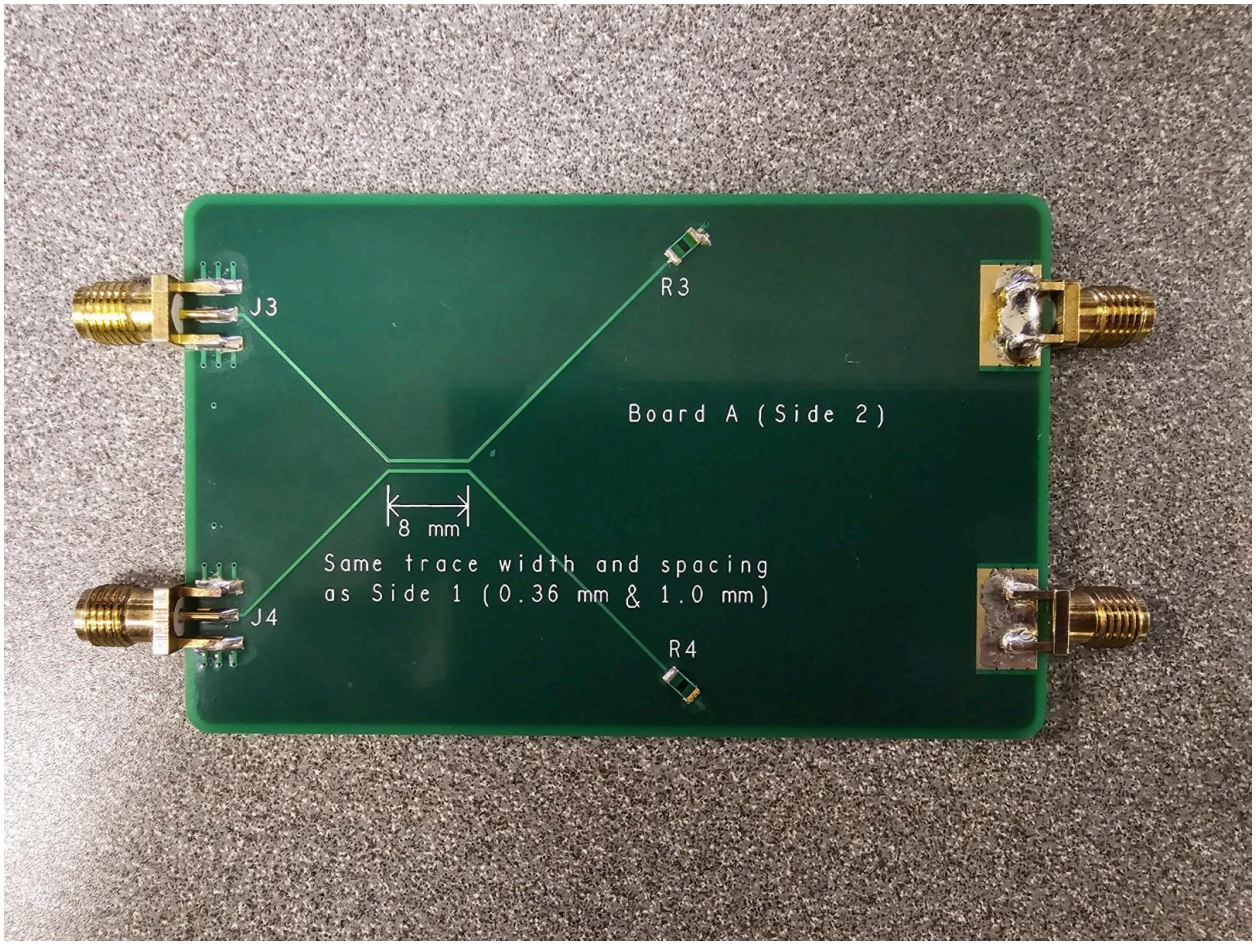
If the traces were spaced farther apart, the coupling would decrease because the electric fields would interact less. The electric field is the primary contributor to coupling in this case.

The S11 plot shows that the circuit is well matched because it remains below –10 dB, meaning most of the signal is absorbed rather than reflected.

When R1 is shorted, the 50 Ω termination is removed, creating an impedance mismatch. This causes part of the electromagnetic wave traveling along the transmission line to reflect back. These reflections interfere with the forward wave, which increases S11 (worse matching) and slightly changes S21.

Shorting R2 has very little effect because it is not directly in the main signal path.

Part 2 – Board A (Side 2)



Objective

The goal is to repeat the measurements on the opposite side of the board and compare coupling behavior.

Theory

Coupling depends on both spacing and how long traces run parallel to each other.

Setup

Board A was flipped to Side 2. Port 1 was connected to J3 and Port 2 was connected to J4 using the same SMA cables. The VNA settings remained unchanged.

The S21 plot was zoomed in, and trace markers were used to identify key values. The display was returned to the full view to analyze S11.

Equations used in each experiment

Same as Part 1

Results/Data

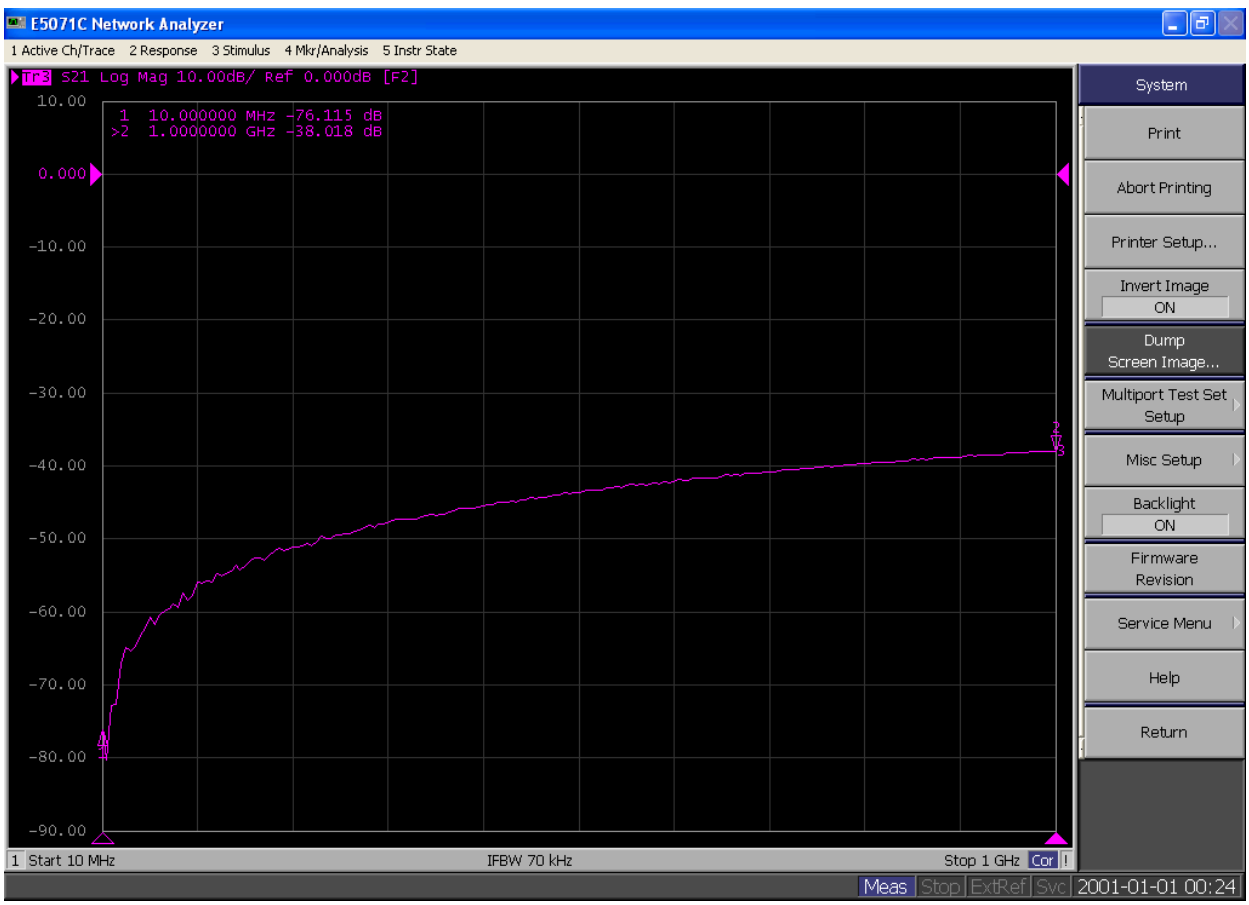


Figure 3. Zoomed S21 plot of Board A, Side 2.

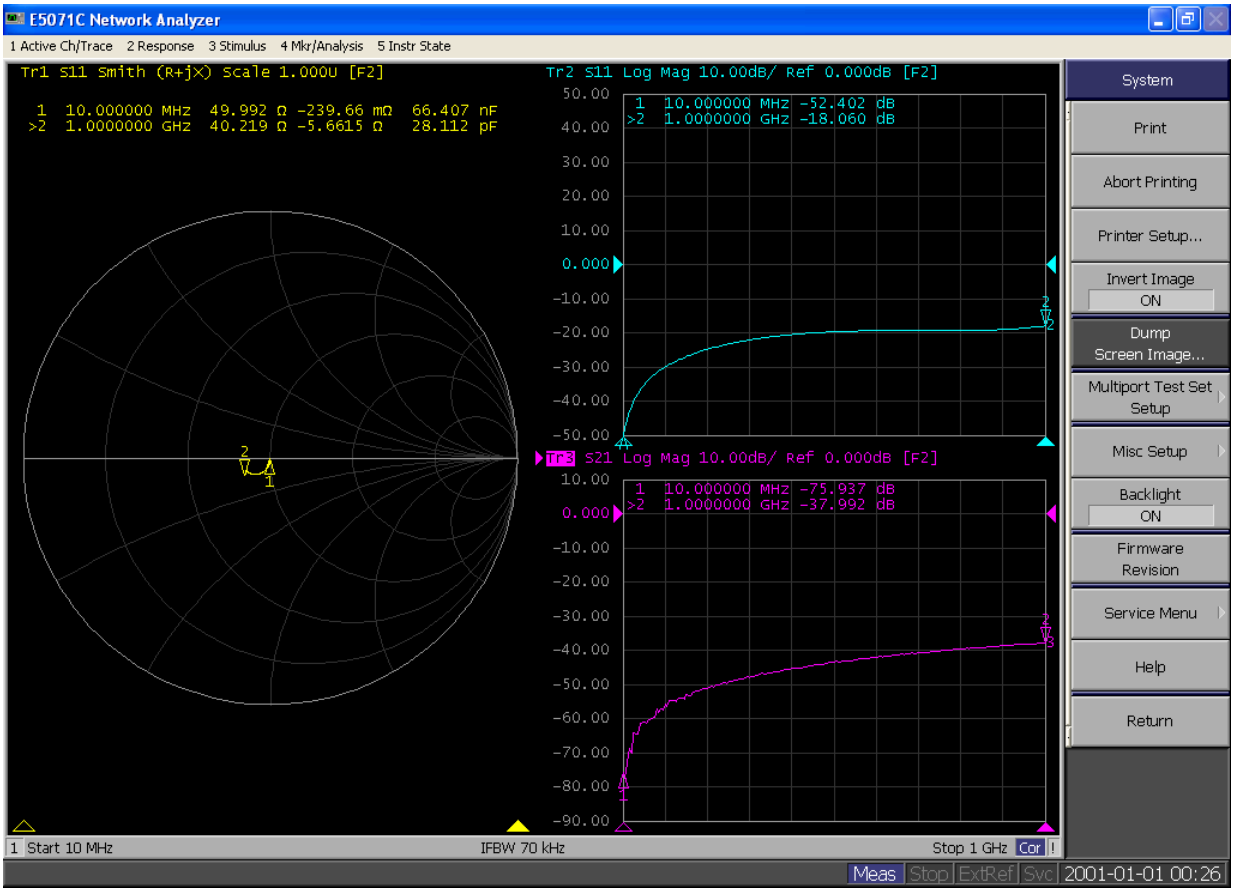


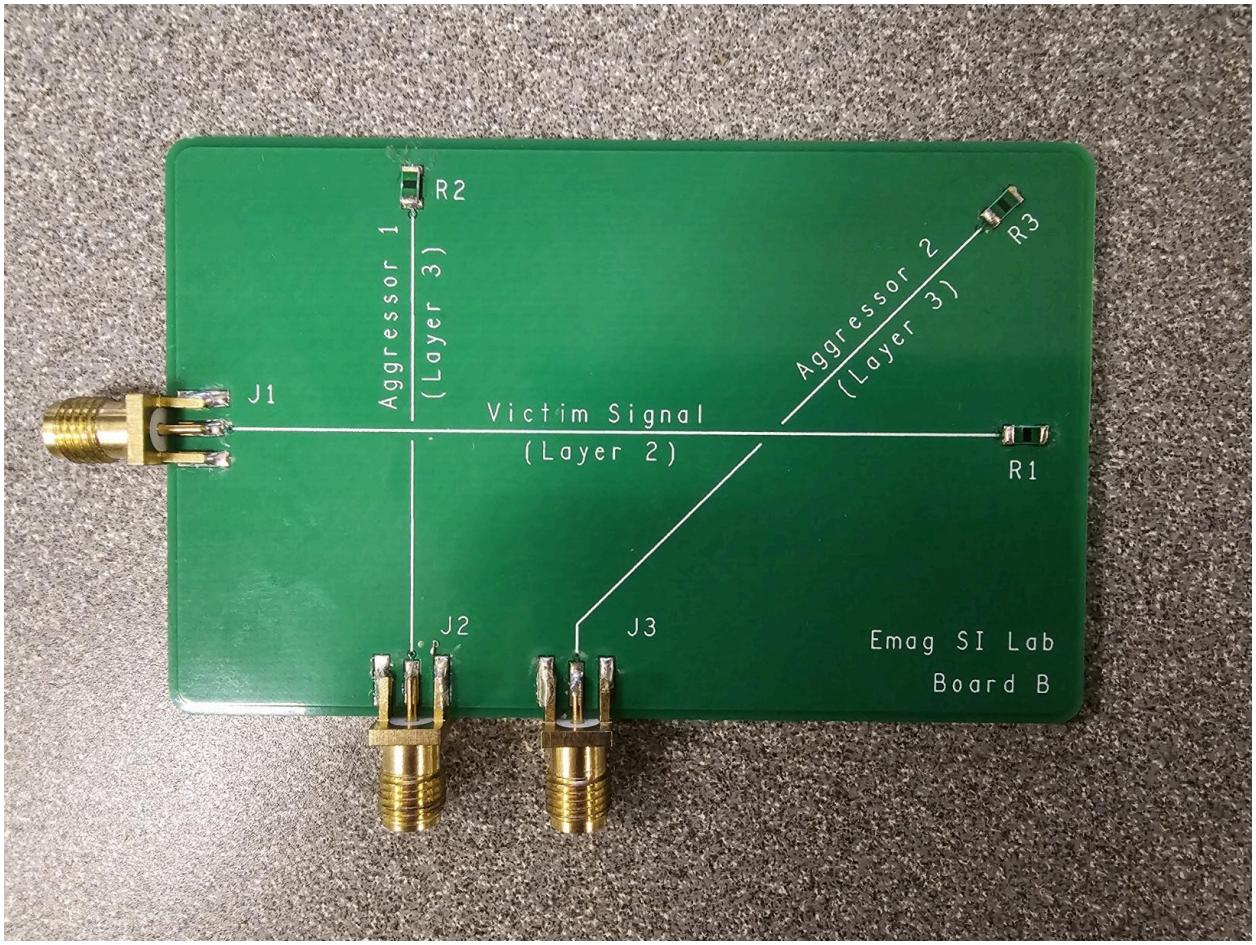
Figure 4. Board A, Side 2 showing S11 and S21 plots.

Discussion

Side 2 shows different coupling compared to Side 1. Even though the spacing between traces is the same, the length over which the traces run parallel is shorter.

This reduces the interaction of the electric fields, which leads to lower coupling. This demonstrates that both spacing and parallel length affect coupling.

Part 3 – Board B (J1 to J2)



Objective

The goal is to measure coupling when traces are on inner layers and routed perpendicular to each other.

Theory

Perpendicular traces reduce coupling because their electric fields do not align, which limits interaction between signals.

Setup

Board B was used for this part. Port 1 of the VNA was connected to J1 and Port 2 was connected to J2. The same VNA settings were used.

The S21 plot was zoomed in, and markers were used to identify the highest and lowest values. The display was returned to the full view to analyze S11. A conductive object was used to short R2, and the changes were observed.

Equations used in each experiment

Same as previous parts

Results/Data

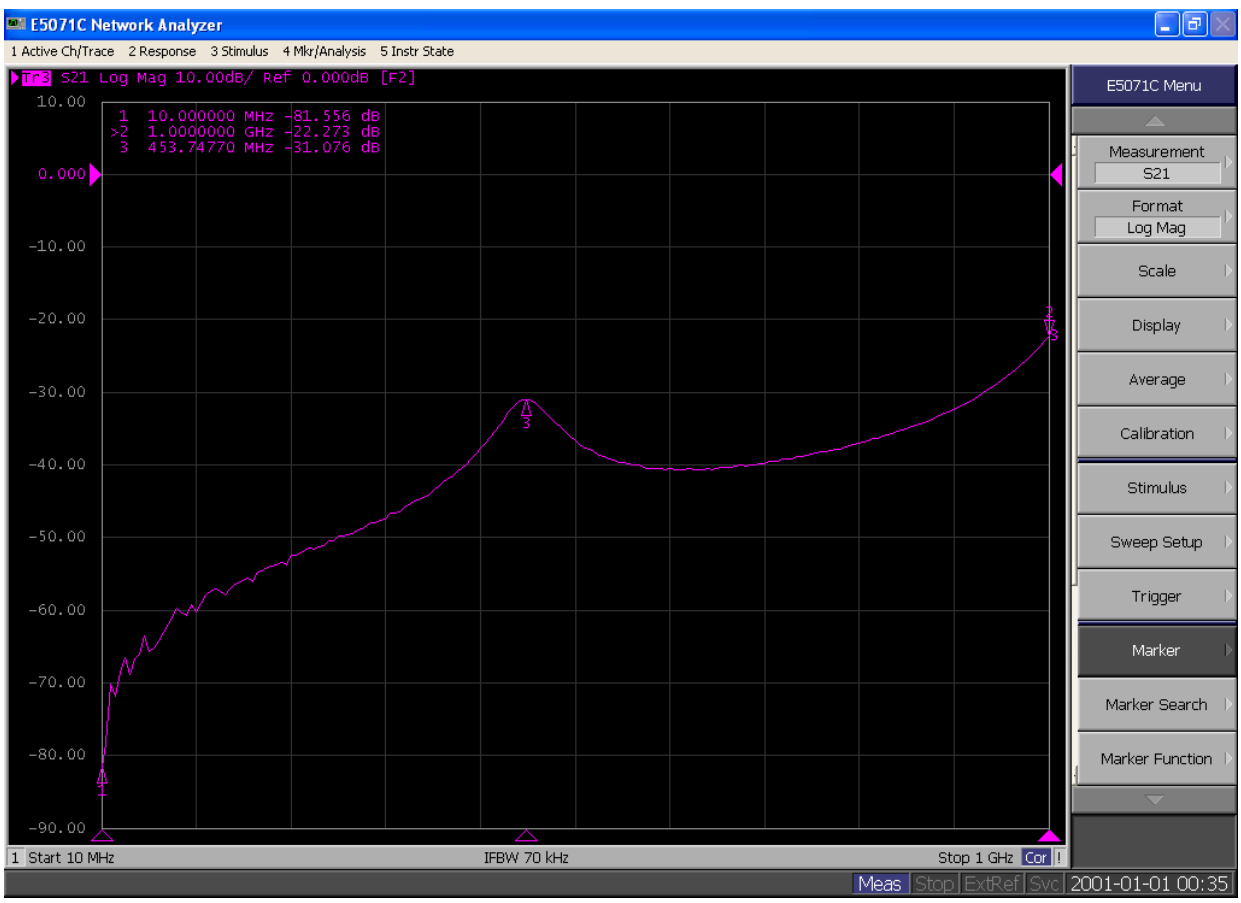


Figure 5. S21 plot of Board B (J1 to J2).

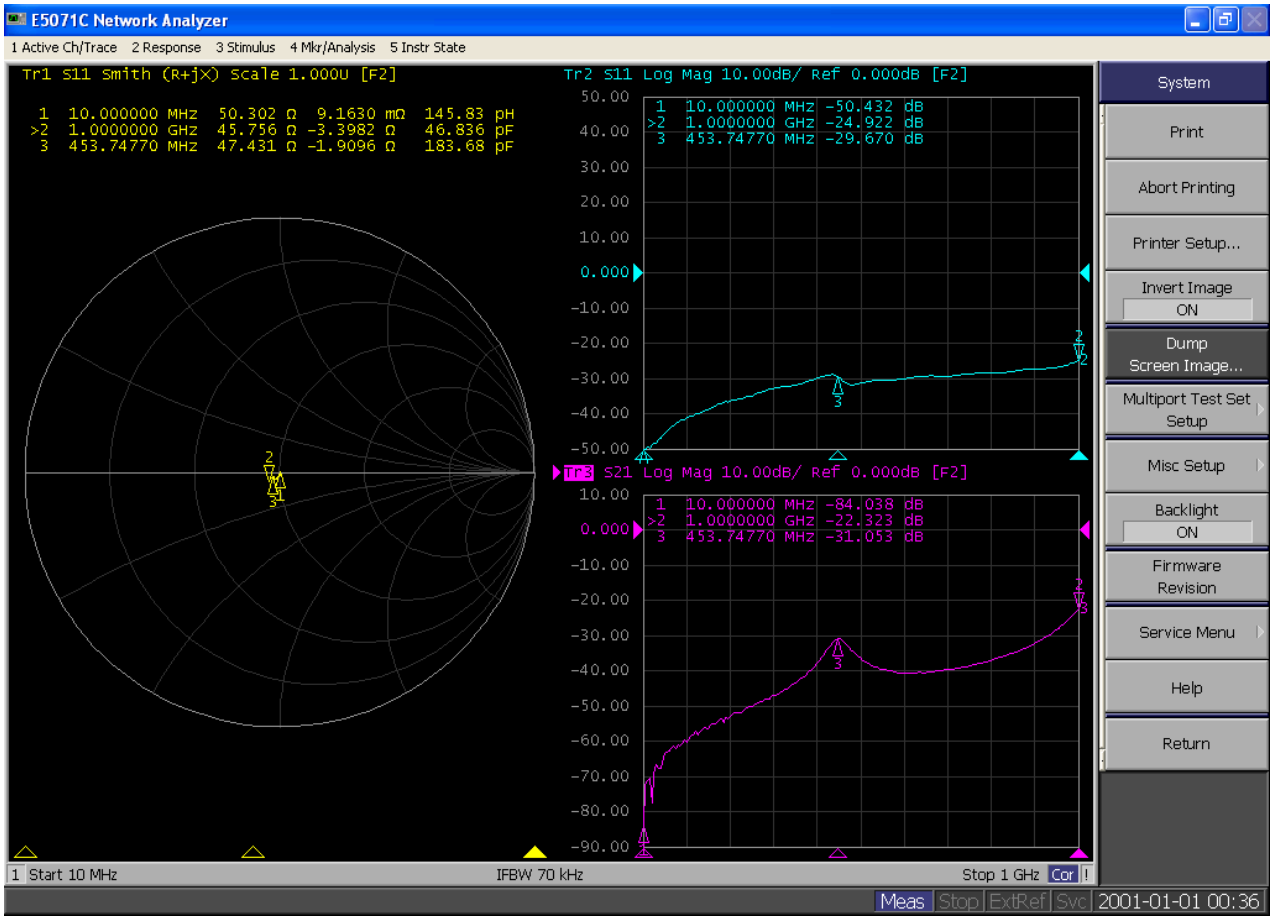


Figure 6. Board B (J1 to J2) showing S11 and S21 plots.

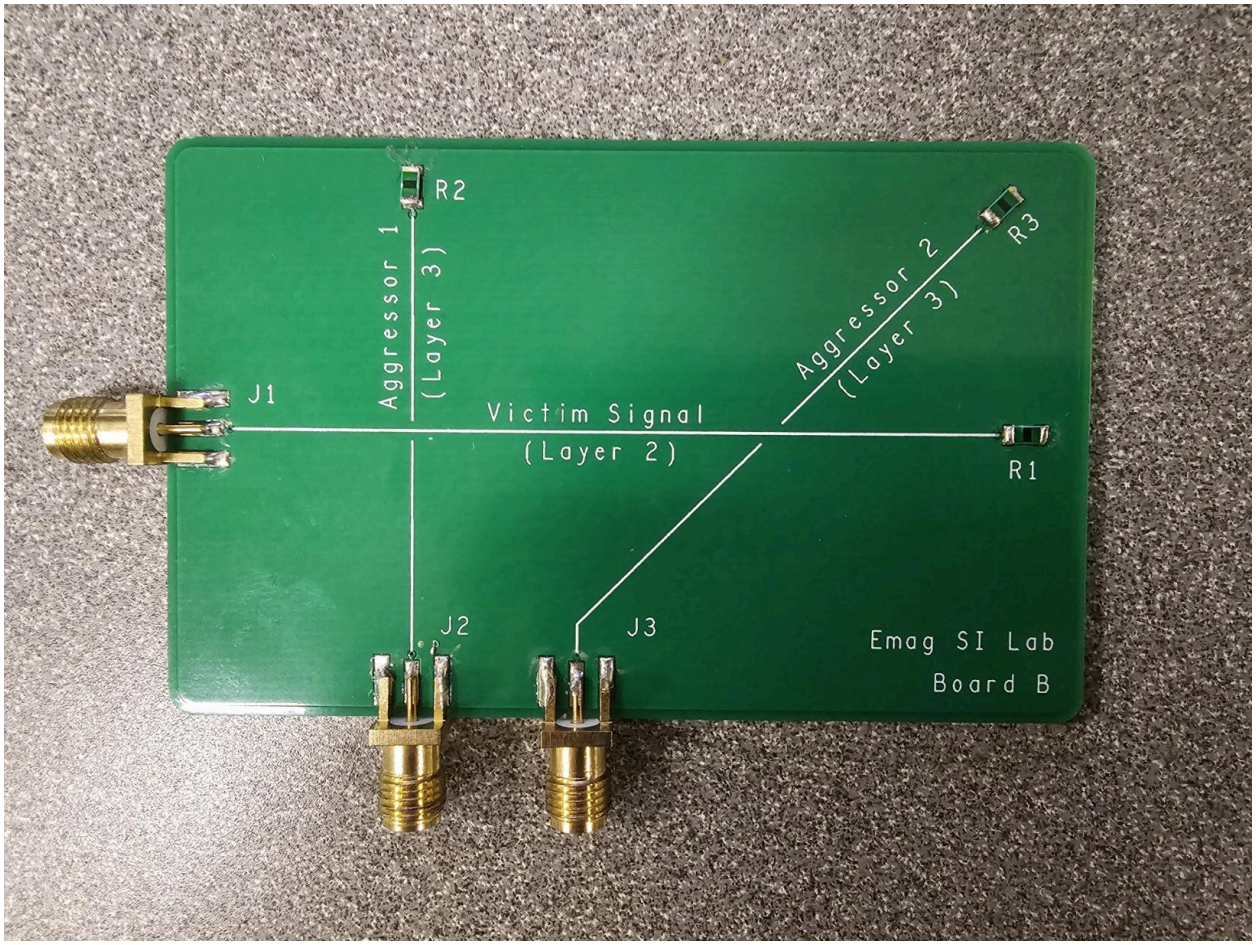
Discussion

The S21 plot shows a peak at a certain frequency, indicating resonant behavior where more signal is transferred.

The S11 plot shows good impedance matching because it remains below –10 dB.

Shorting R2 does not significantly affect S21 because it is not directly in the main signal path between J1 and J2.

Part 4 – Board B (J1 to J3)



Objective

The goal is to compare coupling when traces are angled instead of perpendicular.

Theory

Angled traces allow more electric field interaction, which increases coupling.

Setup

Port 1 remained connected to J1, and Port 2 was moved to J3. The same measurements were repeated.

Equations used in each experiment (if applicable)

Same as previous parts

Results/Data

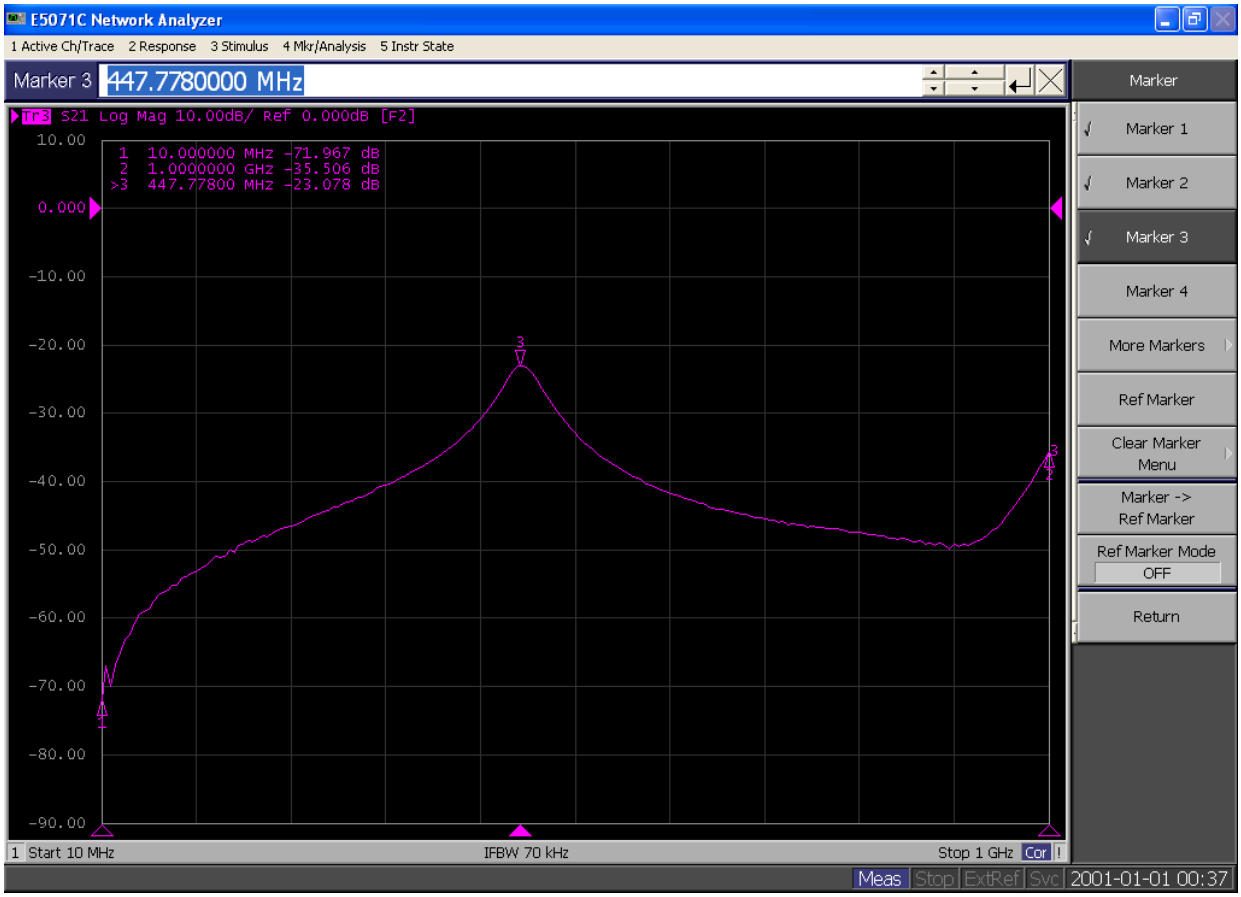


Figure 7. S21 plot of Board B (J1 to J3).

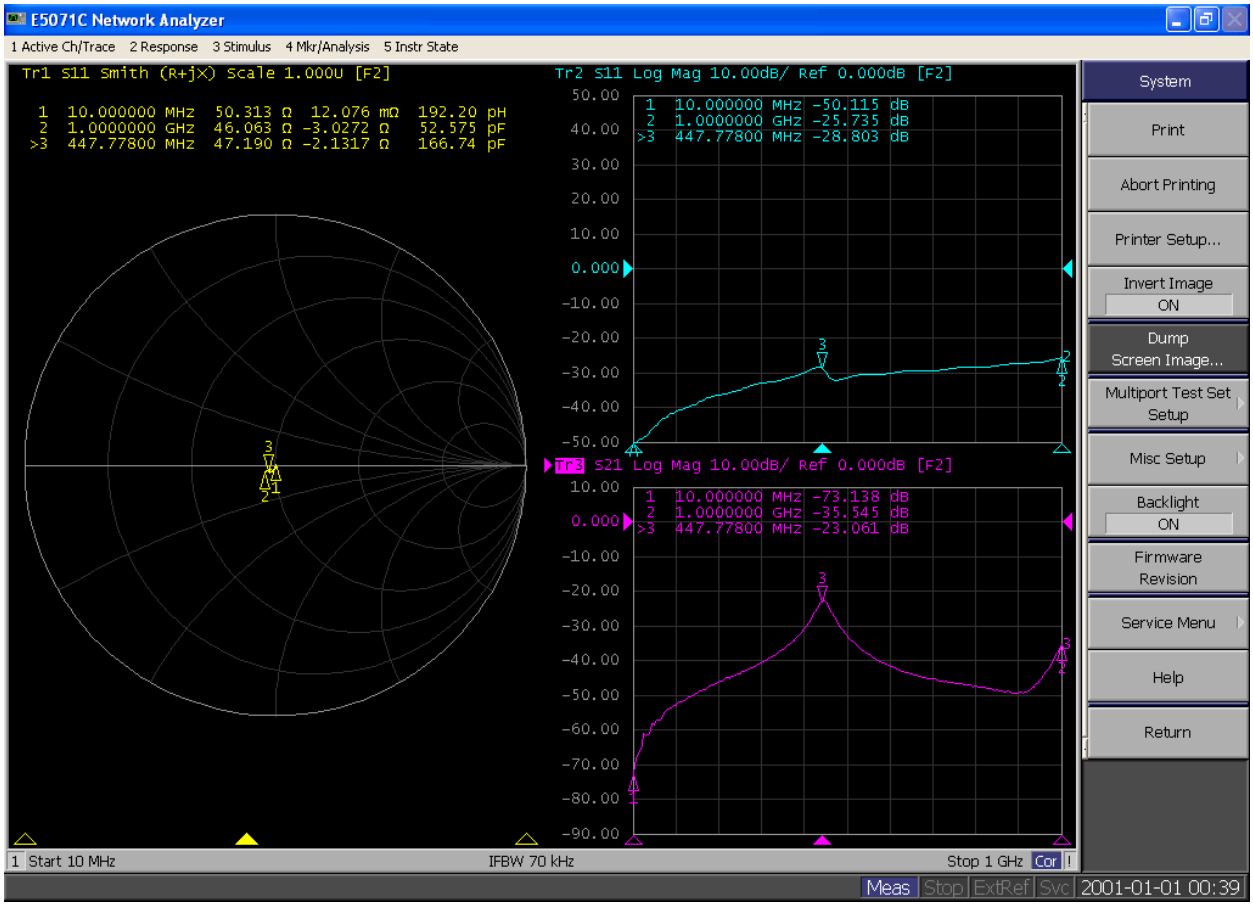


Figure 8. Board B (J1 to J3) showing S11 and S21 plots.

Discussion

Coupling is stronger compared to J1–J2. This is because the traces are angled, allowing more overlap of the electric fields.

Perpendicular routing reduces coupling more effectively than angled routing.

Part 5 – Recalibration (3 GHz)

Objective

The goal is to observe how signal behavior changes at higher frequencies.

Setup

The VNA was recalibrated, and the frequency range was changed to 10 MHz to 3 GHz. The same measurements were repeated for both Board B and Board A.

Equations used in each experiment (if applicable)

Same as previous parts

Results/Data

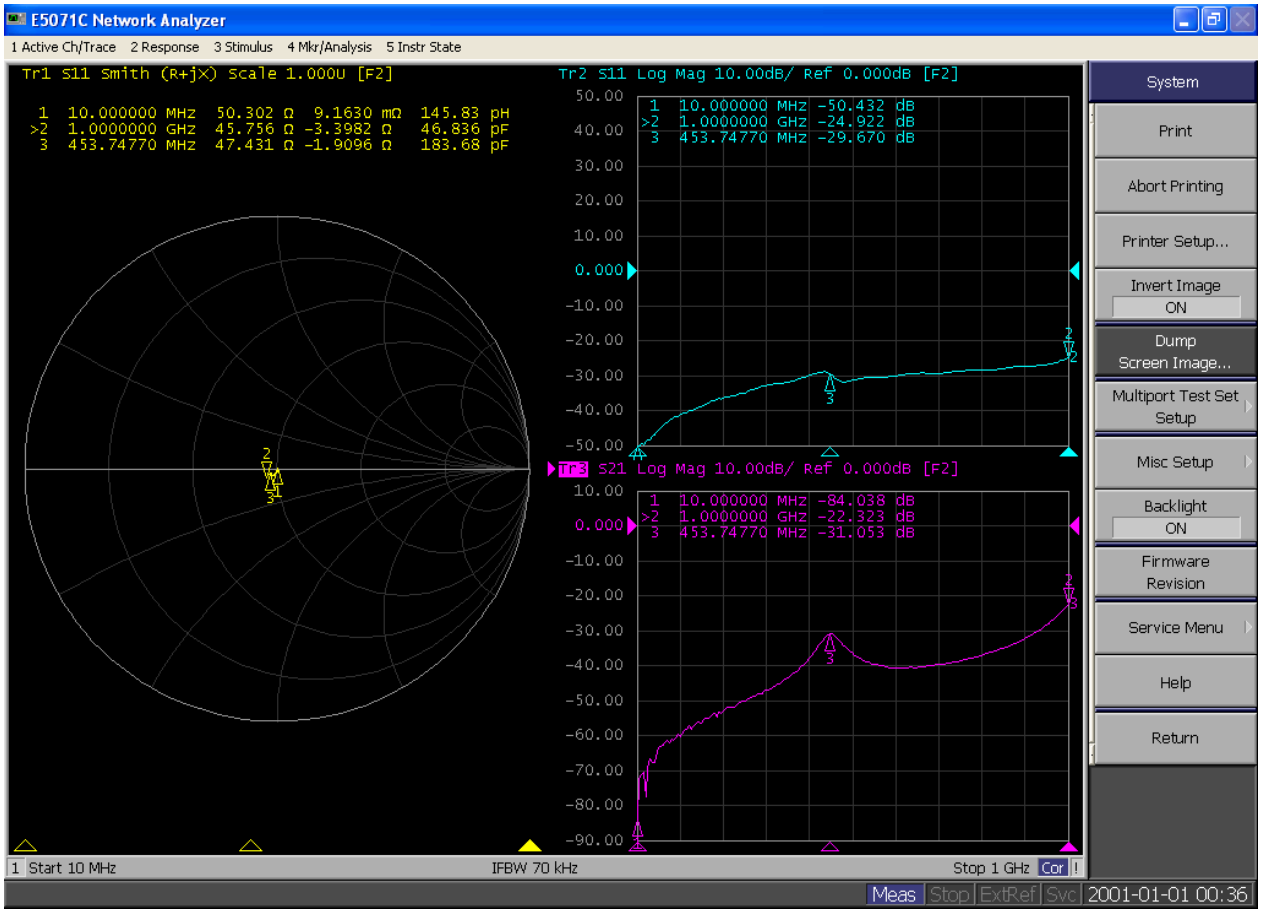


Figure 9. Board B measurement from 10 MHz to 3 GHz.

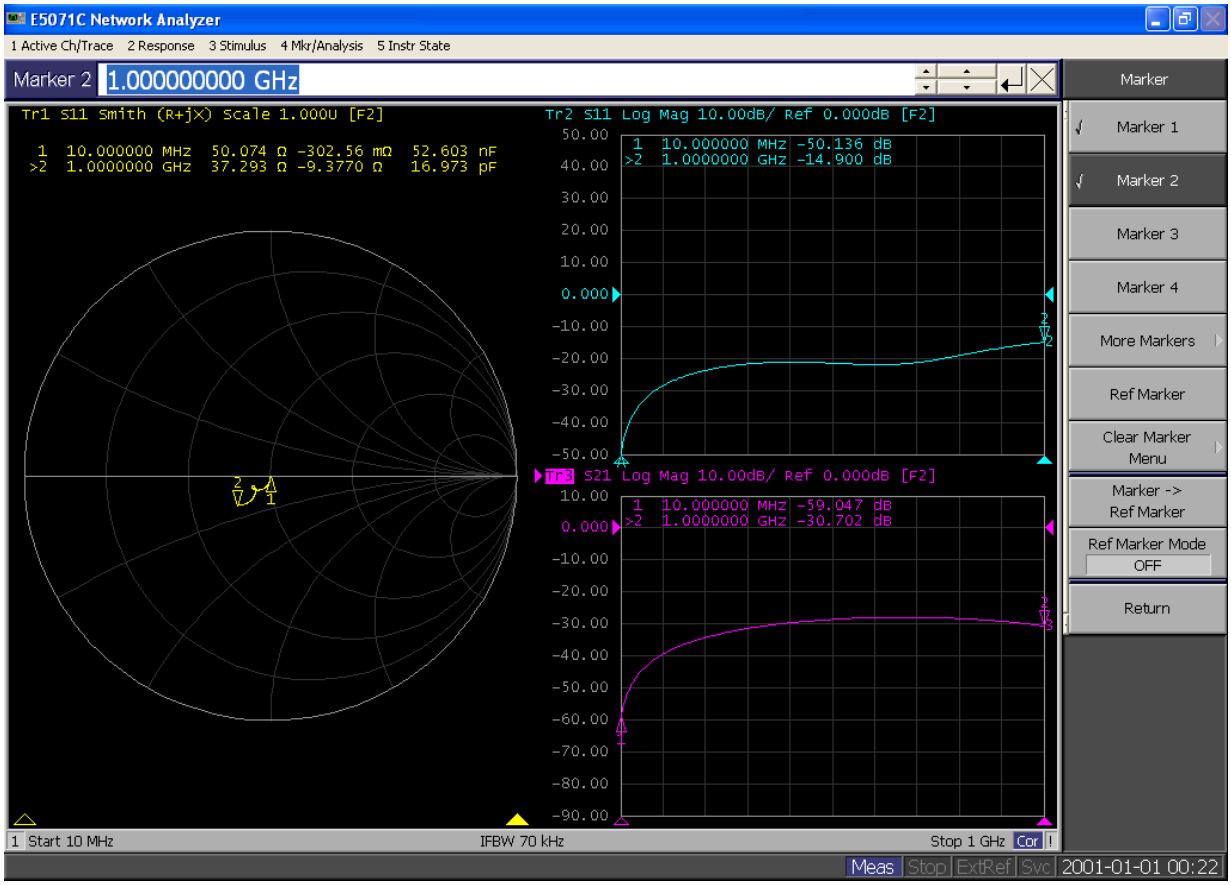


Figure 10. Board A measurement from 10 MHz to 3 GHz.

Discussion

At higher frequencies, signals behave more like waves and become more sensitive to layout. This causes higher variation in both S21 and S11.

Part 6 – Capstone Lab

Objective

The goal is to observe the signal coupling created by two separate traces; this is observed by using a function generator and measuring the coupled signal on the second trace with an oscilloscope.

Setup

Board A was connected to a signal generator on port J1 and an oscilloscope on port J2 using BNC-to-SMA adapters. The signal generator was set to a 5 V peak-to-peak square wave at 100 MHz with a 2.5 V offset.

The output signal was observed on the oscilloscope, and the board was flipped to side B connected to the signal generator on port J3 using the same square wave as side A and an oscilloscope on port J4 and the same measurement was observed.

Equations used in each experiment (if applicable)

None

Results/Data

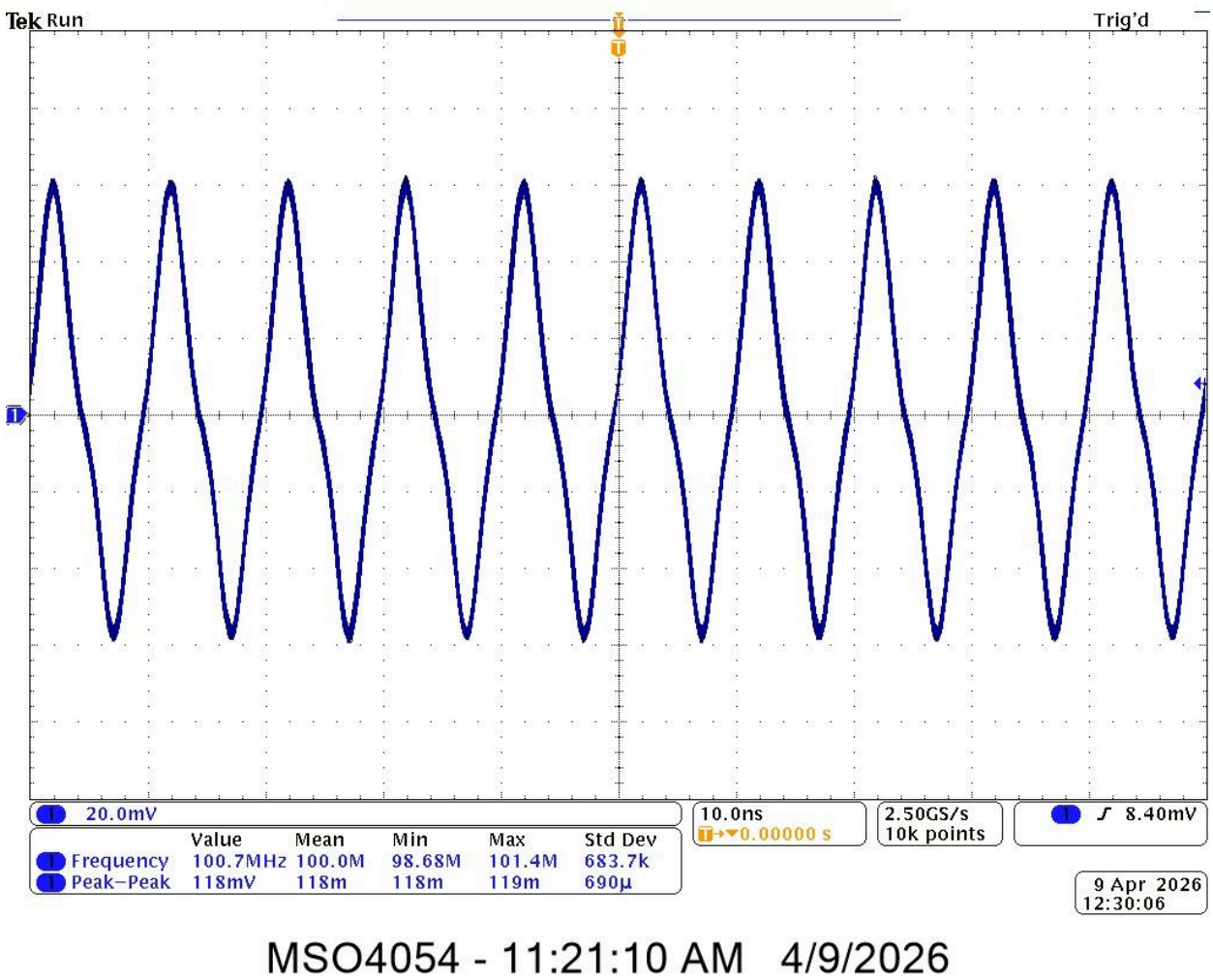


Figure 11. Oscilloscope waveform (Board A Side 1).

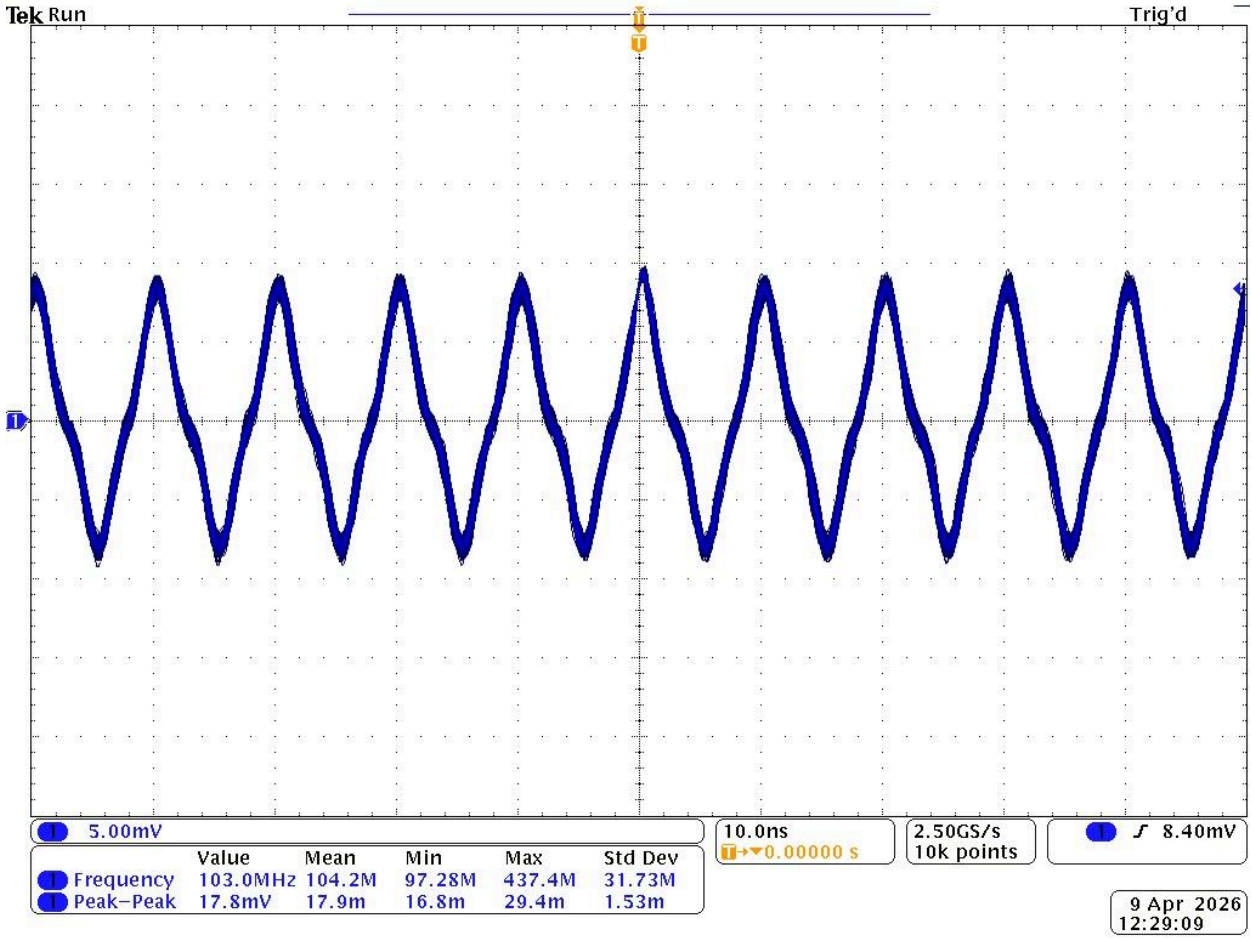


Figure 12. Oscilloscope waveform (Board A Side 2).

Discussion

The oscilloscope is not good for weak signals because it measures the output in the time domain, then uses math to calculate the frequency domain, resulting in lower quality frequency domain output. The oscilloscope requires a stronger input to ensure it has enough data to process. The VNA on the other hand directly measures frequency purity in the frequency domain.

In this lab, a signal generator had to be used to make the coupling visible. The VNA, however, generates its own signal and can measure very small signals across a range of frequencies, making it much more effective for analyzing coupling.

Questions

Part 1

1. What does zooming into S21 mean?
   a. It means focusing on how much signal is transmitted between the two ports, which shows the level of coupling between the traces.
2. Would you want the signals to be highly coupled?
   a. No, for independent signals we want low coupling to avoid interference. High coupling is only desired when transferring power.
3. If the traces were further apart, what would happen?
   a. They would be less coupled because the electric fields would interact less.
4. Which field contributes most to coupling?
   a. The electric field.

Part 2 / Board B

1. Why is coupling different between J1–J2 and J1–J3?
   a. J1–J2 traces are perpendicular, which reduces coupling. J1–J3 traces are angled, allowing more electric field interaction and increasing coupling.

Final Questions

1. Why does S21 change when resistors are shorted?
   a. Shorting removes the 50 Ω termination, causing an impedance mismatch. This leads to reflections of the electromagnetic wave along the transmission line, which interferes with the forward wave and slightly changes S21.
2. Is an oscilloscope a great tool for observing lightly-coupled signals? Explain.
   a. No, as discussed in part 6, the oscilloscope measures in the time domain and then converts into a frequency domain. It's more like an estimation based on visible trace.
3. Why is a VNA better than an oscilloscope?
   a. a. A VNA can detect very small signals and sweep across frequencies automatically, while an oscilloscope requires a strong input signal and cannot easily detect small coupling.
4. How can coupling be reduced?
   a. Increase spacing between traces
   b. Route traces perpendicular to each otheR
   c. Use ground planes
   d. Keep traces as short as possible

- Missing 3 GHz plots. The ones shown are 1 GHz

  Jeff Dinsmore , Apr 27 at 5:05pm

- Magnetic field is the dominant field affecting coupling in traces that are parallel or at a close angle. Electric field may become dominant in high impedance lines but these are low impedance
- On How can coupling be reduced? Report says "Keep traces as short as possible". The correct answer is differential signaling

  Jeff Dinsmore , Apr 28 at 10:58pm